

A CLASS OF DISTRIBUTION-FREE STATISTICS FOR HOMOGENEITY AGAINST ORDERED ALTERNATIVES

Narinder Kumar: Department of Statistics, Panjab University,
Chandigarh, India
Amar Nath Gill: Department of Statistics, Panjab University,
Chandigarh, India
Anil Kumar Dhawan: Centre for Computer Science and Applications,
Panjab University, Chandigarh, India.

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Summary: In this paper a class of distribution-free tests for the problem of testing homogeneity of location parameters of k distributions against the ordered alternatives, based on Hoeffding's (1948) U-statistic, is proposed. The asymptotic distribution of the test statistic under the equality of location parameters as well as under the equally spaced sequence of local alternatives is derived. Pitman asymptotic relative efficiencies of the members of the proposed class of tests relative to existing tests are obtained, with interesting results.

1. INTRODUCTION

Let $X_{i1}, \dots, X_{in}$ be a random sample of size n from the absolutely continuous distribution with cumulative distribution function (cdf) $F_i(x) = F(x - \theta_i)$, $i = 1, \dots, k$, where $F(\cdot)$ is any absolutely continuous cdf. Assume that the F_i are identical in all respects except for their location parameters. We want to test the null hypothesis of homogeneity of location parameters, that is,

$$H_0 : \theta_1 = \theta_2 = \dots = \theta_k$$

against the ordered alternatives

$$H_A : \theta_1 \leq \theta_2 \leq \dots \leq \theta_k$$

with at least one strict inequality.

For some earlier work on this problem reference may be made to Jonckheere (1954), Tryon and Hettmansperger (1973), Govindarajulu and Haller (1977), Amita and Kochar (1989), among others. A variety of statistical inference problems under order restrictions can be found in Barlow et al. (1972), and Puri and Sen (1971). For testing H_0 against H_A , we propose a class of distribution-free tests, based on weighted linear combinations of consecutive two-sample U-statistics. The tests are given in section 2 and their distributions are discussed in section 3. In section 4, optimal weighting coefficients which maximize the efficacy are determined when the location parameters are assumed to be equally spaced. The Pitman asymptotic relative efficiencies of the proposed tests, with the optimal weighting coefficients, relative to the tests of Jonckheere (1954), and Amita and Kochar (1989) are given in section 5.

2. THE PROPOSED TESTS

For $i < j$, under the alternative hypothesis H_A , the observations from the i -th population are expected to be smaller than those from the j -th population. First we propose a two-sample U-statistic in the context of a two-sample location problem and then treat the k -sample problem.

For a common sample of size $n \geq 3$ from each distribution, define a positive integer m such that

$$1 \leq m \leq \frac{n}{2} - 1 \quad \text{for } n \text{ even}$$

and

$$1 \leq m \leq \frac{n-1}{2} \quad \text{for } n \text{ odd.}$$

Now for $i < j$, $i, j=1, \dots, k$, define the kernel

$$\Phi(X_{i\alpha_1}, \dots, X_{i\alpha_{2m+1}}; X_{j\beta_1}, \dots, X_{j\beta_{2m+1}}) = \begin{cases} 1 & \text{if } \text{Med}(X_{i\alpha_1}, \dots, X_{i\alpha_{2m+1}}) \leq \text{Med}(X_{j\beta_1}, \dots, X_{j\beta_{2m+1}}) \\ 0 & \text{otherwise.} \end{cases}$$

The two-sample U-statistic U_{ij}^m corresponding to the above kernel is given by

$$U_{ij}^m = \left[\binom{n}{2m+1} \right]^{-2} \sum_0 \Phi(X_{i\alpha_1}, \dots, X_{i\alpha_{2m+1}}; X_{j\beta_1}, \dots, X_{j\beta_{2m+1}}),$$

where $\sum_0$ denotes the sum over all combinations $(\alpha_1, \dots, \alpha_{2m+1})$ of $(2m+1)$ integers chosen from $(1, \dots, n)$ and all combinations $(\beta_1, \dots, \beta_{2m+1})$ of $(2m+1)$ integers chosen from $(1, \dots, n)$. Clearly, for $m = 1$, U_{ij}^m is the statistic proposed by Shetty and Govindarajulu (1988) for the two-sample location problem and later used by Kumar et al. (1994) in the context of ordered alternatives.

Motivated by the fact that under H_A , $U_{i,i+1}^m$, $i=1, \dots, k$ is expected to take large values, we propose the class of statistic

$$W_m = \sum_{i=1}^{k-1} a_i U_{i,i+1}^m$$

for testing H_0 against H_A , where $a_1, \dots, a_{k-1}$ are suitably chosen real constants. It may be noted that for each value of m , in its domain of definition, W_m is a member of the proposed class and each member gives a class of test statistics for variations in vector $(a_1, \dots, a_{k-1})$. A large value of W_m leads to the rejection of H_0 .

3. DISTRIBUTION OF W_m

$$\text{Clearly, } E(W_m) = \sum_{i=1}^{k-1} a_i \mu_{i,i+1}^m,$$

$$\text{where } \mu_{i,i+1}^m = E(U_{i,i+1}^m)$$

$$= \frac{(2m+1)!}{(m!)^2} \sum_{t=m+1}^{2m+1} \int_{-\infty}^{\infty} (F_i(x))^t (1-F_i(x))^{2m+1-t} (F_{i+1}(x))^m (1-F_{i+1}(x))^m dF_{i+1}(x).$$

Under H_0 ,

$$\begin{aligned} E(W_m) &= \left[\frac{(2m+1)!}{(m!)^2} \sum_{t=m+1}^{2m+1} \int_{-\infty}^{\infty} (F(x))^{m+t} (1-F(x))^{3m+1-t} dF(x) \right] \sum_{i=1}^{k-1} a_i \\ &= \left[\frac{(2m+1)!}{(m!)^2} \sum_{t=m+1}^{2m+1} \beta(m+t+1, 3m+2-t) \right] \sum_{i=1}^{k-1} a_i \end{aligned}$$

$$= \frac{1}{2} \sum_{i=1}^{k-1} a_i, \text{ for every } m,$$

where

$$\beta(p, q) = \int_0^1 y^{p-1} (1-y)^{q-1} dy.$$

Consider the random vector $\underline{U}^m = (U_{12}^m, U_{23}^m, \dots, U_{k-1,k}^m)$. Hereafter we omit the superscript in the vector $\underline{U}^m$ for notational convenience. The asymptotic normality of $\underline{U}$, as stated in the following theorem, follows immediately from Lehmann (1963).

THEOREM 3.1: The asymptotic distribution of $N^{\frac{1}{2}}[\underline{U} - E(\underline{U})]$ as $N \rightarrow \infty$ in such a way that $n/N \rightarrow p$, $0 < p < 1$, is multivariate normal with mean vector $\underline{0}$ and dispersion matrix $\Sigma = ((\sigma_{ij}))$, where $N = nk$ and

$$\sigma_{ij} = \begin{cases} \frac{(2m+1)^2}{p} \zeta_{i,i+1;i,i+1}^{(i)} + \frac{(2m+1)^2}{p} \zeta_{i,i+1;i,i+1}^{(i+1)} & \text{for } i=j=1, \dots, k-1 \\ \frac{(2m+1)^2}{p} \zeta_{i,i+1;i+1,i+2}^{(i+1)} & \text{for } j=i+1, i=1, \dots, k-2 \\ \frac{(2m+1)^2}{p} \zeta_{i-1,i;i,i+1}^{(i)} & \text{for } j=i-1, i=2, \dots, k-1 \\ 0 & \text{otherwise,} \end{cases}$$

where

$$\zeta_{i,i+1;i,i+1}^{(i)} = E[\{\varphi_{i,i+1}^{(i)}(X)\}^2] - E^2[U_{i,i+1}^m],$$

$$\zeta_{i,i+1;i,i+1}^{(i+1)} = E[\{\varphi_{i,i+1}^{(i+1)}(X)\}^2] - E^2[U_{i,i+1}^m],$$

$$\zeta_{i,i+1;i+1,i+2}^{(i+1)} = E[\varphi_{i,i+1}^{(i+1)}(X) \varphi_{i+1,i+2}^{(i+1)}(X)] - E[U_{i,i+1}^m]E[U_{i+1,i+2}^m],$$

$$\varphi_{i,j}^{(i)}(x) = E[\Phi_{ij}^m(x, X_{i2}, \dots, X_{i,2m+1}; X_{j1}, \dots, X_{j,2m+1})],$$

$$\varphi_{i,j}^{(j)}(x) = E[\Phi_{ij}^m(X_{i1}, \dots, X_{i,2m+1}; x, X_{j2}, \dots, X_{j,2m+1})].$$

Under H_0 , dispersion matrix Σ becomes $\Sigma^* = ((\sigma_{ij}^*))$, where

$$(3.1) \quad \sigma_{ij}^* = \begin{cases} 2k(2m+1)^2 \rho_m & \text{for } i=j=1, \dots, k-1 \\ -2k(2m+1)^2 \rho_m & \text{for } j=i+1, i=1, \dots, k-2 \\ -2k(2m+1)^2 \rho_m & \text{for } j=i-1, i=2, \dots, k-1 \\ 0 & \text{otherwise,} \end{cases}$$

$p = 1/k$ and

$$\begin{aligned} \rho_m = (2m+1)^2 \left[\binom{2m}{m} \right]^4 \sum_{j=0}^{2m} \sum_{j'=0}^{2m} \frac{(-1)^{j+j'} \binom{2m}{j} \binom{2m}{j'}}{(2m+j+1)} & \left[\frac{1}{(2m+j'+2)} - \right. \\ & \left. \frac{1}{(2m+j+2)(4m+j+j'+3)} \right] \\ & + (2m+1)^2 \left[\binom{2m}{m} \right]^2 \left[\sum_{i=1}^m \binom{2m}{m+i} \beta(2m+1+i, 2m+1-i) \right]^2 \\ & + 2(2m+1)^2 \left[\binom{2m}{m} \right]^3 \left[\sum_{i=1}^m \binom{2m}{m+i} \beta(2m+1+i, 2m+1-i) \right] \left[\sum_{j=0}^{2m} \frac{(-1)^j \binom{2m}{j}}{(2m+j+2)} \right] - \frac{1}{4}. \end{aligned}$$

Here $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

Since W_m is a linear combination of the components of $\underline{U}$, the proof of the following theorem follows from the transformation theorem (Serfling 1980, p. 122).

THEOREM 3.2: The asymptotic distribution of $N^{\frac{1}{2}}[W_m - E(W_m)]$ as $N \rightarrow \infty$ is normal with mean zero and variance $\underline{a}' \Sigma \underline{a}$, where $\underline{a}' = (a_1, \dots, a_{k-1})$. Under H_0 ,

$$E(W_m) = \frac{1}{2} \sum_{i=1}^{k-1} a_i$$

and

$$\underline{a}' \Sigma \underline{a} = 2k(2m+1)^2 \rho_m \left[\sum_{i=1}^{k-1} a_i^2 - \sum_{i=1}^{k-2} a_i a_{i+1} \right].$$

4. OPTIMAL CHOICE OF WEIGHTS

In this section, we consider the problem of obtaining optimal weights a_i , so that for fixed m the test W_m has maximum efficacy for the sequence of Pitman type alternatives

$$(4.1) \quad H_N : F_i(x) = F(x - N^{-\frac{1}{2}}\theta_i), \quad i=1, \dots, k.$$

For efficiency comparisons, we consider the equally spaced alternatives of the type $\theta_i = i\theta$, $\theta > 0$ for $i=1, \dots, k$. Then the alternative H_N becomes

$$(4.2) \quad H'_N : F_i(x) = F(x - N^{-\frac{1}{2}}i\theta), \quad i=1, \dots, k.$$

The following theorem gives the asymptotic distribution of $\underline{U}$ under the sequence of local alternatives $\{H'_N\}$. The proof is routine and hence omitted.

THEOREM 4.1: Let X_{ij} be independent random variables with cumulative distribution functions $F_i(x)$, $i=1, \dots, k$, $j=1, \dots, n$, where the F_i are given by (4.2). The limiting distribution of $N^{\frac{1}{2}}[\underline{U} - (\frac{1}{2})E_{k-1}]$ is $(k-1)$ -dimensional multivariate normal with mean vector μE_{k-1} and dispersion matrix $\Sigma^* = ((\sigma_{ij}^*))$, where

$$\mu = (2m+1)(m+1) \binom{2m}{m} \binom{2m+1}{m+1} \theta \int_{-\infty}^{\infty} (F(x))^{2m} (1-F(x))^{2m} f^2(x) dx$$

$$\text{and } E_{k-1} = [1]_{(k-1) \times 1}.$$

Proceeding as in Govindarajulu and Gupta (1978), we can easily prove the following theorem.

THEOREM 4.2: Under the sequence $\{H'_N\}$ given by (4.2), the efficacy of the test W_m is maximized when

$$a_i = \frac{i(k-1)}{2k}, \quad i=1, \dots, k-1.$$

The efficacy of the U_m test with this optimum choice of weights is

$$(4.3) \quad e(W_m) = \frac{(k^2-1)(m+1)^2 \left(\frac{2m}{m}\right)^2 \left(\frac{2m+1}{m+1}\right)^2 G_m^2}{12\rho_m},$$

where

$$G_m = \int_{-\infty}^{\infty} (F(x))^{2m} (1-F(x))^{2m} f^2(x) dx.$$

Clearly, $e(W_m)$ depends on m .

5. ASYMPTOTIC RELATIVE EFFICIENCIES

In this section, we compare the asymptotic relative efficiencies of W_m tests relative to some other known nonparametric tests for this testing problem.

The Pitman asymptotic relative efficiency of the W_m test with respect to Jonckheere's test (J) is

$$(5.1) \quad e(W_m, J) = \frac{(m+1)^2 \left(\frac{2m}{m}\right)^2 \left(\frac{2m+1}{m+1}\right)^2 G_m^2}{12\rho_m} \left[\int_{-\infty}^{\infty} f^2(x) dx \right]^{-2}.$$

Amita and Kochar (1989) also proposed a class of distribution-free tests (L_s) for the problem considered here. The efficacy of their test with this optimal choice of weights is

$$(5.2) \quad e(L_s) = \frac{\left[\int_{-\infty}^{\infty} [(F(x))^{s-2} + (\bar{F}(x))^{s-2}] f^2(x) dx \right]^2 (k^2-1)}{24[(s-1)^2(2s-1)]^{-1} [1 - \left(\frac{2s-2}{s-1}\right)^{-1}]},$$

where s is the sum of the sub-sample sizes taken from the two samples.

From (4.3) and (5.2), we find that asymptotic relative efficiency of W_m tests relative to L_s tests is

$$(5.3) \quad e(W_m, L_s) = \frac{e(W_m)}{e(L_s)}$$

The AREs in (5.1)((5.3)) computed for various choices of m and s and for some common distributions are given in table I(II).

Table I. AREs of W_m test relative to J test

Distribution m	1	2	3
Uniform	0,588	0,511	0,472
Logistic	0,972	0,949	0,926
Double Exp.	1,111	1,160	1,180
Exponential	0,588	0,511	0,472

Table II. AREs of W_m test relative to L_s test

Distribution s		4	5	6	7	8	9
Double Exp.	m						
	1	1,182	1,298	1,447	1,619	1,805	2,000
	2	1,234	1,355	1,511	1,691	1,885	2,089
Logistic	m						
	1	0,977	0,988	1,035	1,085	1,142	1,206
	2	0,954	0,975	1,011	1,059	1,115	1,178
Uniform	m						
	1	0,479	0,387	0,319	0,271	0,235	0,208
	2	0,416	0,336	0,277	0,235	0,204	0,181
Exponential	m						
	1	0,479	0,387	0,319	0,271	0,235	0,208
	2	0,416	0,336	0,277	0,235	0,204	0,181
	3	0,385	0,311	0,256	0,218	0,189	0,149

On the basis of the distributions considered in the ARE comparisons, we recommend the use of the proposed tests for heavy-tailed distributions as

compared to the tests of Jonckheere, and Amita and Kochar. Moreover, the AREs of our tests as compared to the tests of Amita and Kochar increase as s increases for the moderate and heavy tailed distributions but decreases as s increases for light tailed distributions. This unusual behaviour in AREs, for variations in s , could not be predicted mathematically on the basis of the tail of a distribution.

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